In Silico 4D-CT-Based Tumor Motion Modeling and Quantitative Evaluation of PTV Margins for Lung Radiotherapy Planning

Vu Minh Anh*

Faculty of Information Technology, Ho Chi Minh City University of Industry and Trade (HUIT), Ho Chi Minh City, Vietnam

*Corresponding Author: mvu103536@gmail.com | Target Journal: *Biomedical Signal Processing and Control* (Elsevier)

Abstract: Target motion during thoracic external beam radiotherapy induces spatial misalignment between planned radiation apertures and mobile tumor volumes. While commercial treatment planning systems (TPS) offer sophisticated tools, uniform isotropic planning target volume (PTV) margin expansions remain widely utilized in standard clinical workflows due to practical simplicity. In this in silico investigation, we developed an open-source computational Python pipeline to track 3D tumor centroids across ten-phase 4D-CT volumes and quantitatively evaluate PTV margin strategies using a calibrated synthetic digital thorax phantom. A spatiotemporal volume of interest (VOI, 140 × 135 × 135 mm^3, voxel lattice 2.0 × 1.5 × 1.5 mm^3) was formulated across phases T0 to T9. Respiration kinematics incorporated an asymmetric 60:40 exhalation:inhalation dwell ratio, a generalized Lujan model with power coefficient n = 2, and phase-lag hysteresis (ϕ = 36°) between craniocaudal and anteroposterior trajectories. Phase T4 was designated as the end-exhalation reference, where the cosine-power model reaches its displacement minimum ($\cos^4(\pi/2) = 0$). Lesion volumes were segmented via automated 4-step thresholding with 3D morphological hole-filling, achieving sub-voxel centroid tracking accuracy (3D Mean Absolute Error MAE < 0.035 mm vs. analytical ground truth). We evaluated three margin design strategies: (1) uniform isotropic expansions (2 to 12 mm); (2) internal target volume (ITV) plus isotropic setup margins (2 to 5 mm); and (3) mid-position-based anisotropic PTV margins. Target coverage rates (CR) were benchmarked against the analytical van Herk population recipe ($M = 2.5\Sigma + 0.7\sigma$) and ICRU geometric consensus criteria ($CR_{min} \geq 95\%$). Peak-to-peak excursions were dominated by craniocaudal motion: 12.00 mm (regular), 22.00 mm (deep), and 14.67 ± 1.14 mm (irregular, Monte Carlo N=100). The van Herk formula predicted required craniocaudal margins of 8.37 mm (regular) and 10.78 mm (deep)—substantially larger than the clinical 5 mm margin, consistent with the observed empirical CR failure (CR_{min} = 64.7% regular, 11.7% deep). Note that this formula carries its own distributional assumptions and serves as a population-level benchmark. A 12 mm isotropic expansion restored coverage (CR_{min} = 100% in regular) but inflated PTV volume by +182% (from 20.6 to 58.0 cc). In contrast, an ITV with a 3 mm setup margin (23.9 cc) achieved CR_{min} = 100% under both regular and deep breathing. An anisotropic PTV (19.3 cc) achieved CR_{min} = 98.6% under regular breathing; however, under deep breathing (A_z = 22 mm), CR_{min} = 89.0%, falling below the 95% threshold—suggesting ITV-based planning is superior for large-amplitude respiratory motion. Both motion-adapted strategies reduced PTV volume by 59% to 67% compared to the 12 mm buffer (a geometric proxy for normal tissue dose reduction; 4D dose accumulation is needed to confirm dosimetric benefit). These findings provide computational confirmation that uniform 3D margins are geometrically suboptimal for mobile lung tumors with excursion > 5 mm, supporting 4D-CT ITV planning in accordance with AAPM Task Group 76 guidelines.

Keywords: 4D-CT, Lung Radiotherapy, Intrafraction Motion, In Silico Modeling, Centroid Tracking, PTV Margins, van Herk Recipe, AAPM TG-76.

1. Introduction

Geometric accuracy in thoracic external beam radiotherapy is fundamentally compromised by intrafractional respiratory motion. Contraction of the diaphragm and rhythmic chest wall excursions induce substantial time-dependent displacements of lung parenchyma and embedded lesions. As detailed in the American Association of Physicists in Medicine (AAPM) Task Group 76 report [1], tumor motion exceeding 5 mm along any Cartesian axis warrants explicit motion management to mitigate normal tissue toxicity and prevent target underdosing. In stereotactic body radiation therapy (SBRT), where ablative fractional doses often exceed 10 Gy, unmanaged target displacement leads to steep dosimetric penalties at beam boundaries [2, 3].

Conventional static three-dimensional computed tomography (3D-CT) acquisition captures an arbitrary snapshot of breathing anatomy, resulting in motion-blurring artifacts, artificial volume distortion, and significant misestimation of gross tumor volume (GTV) boundaries [4, 5]. Four-dimensional computed tomography (4D-CT) resolves this temporal ambiguity by sorting projection data into discrete respiratory phase bins (typically 10 phases from peak inhalation T0 to end-exhalation T4/T5) correlated with an external respiratory surrogate [6, 7]. Advanced 4D-CT reconstruction strategies, including maximum intensity projection (MIP), allow clinicians to delineate an internal target volume (ITV) encompassing full

respiratory excursion across the breathing cycle [8, 9].

Despite the clinical availability of 4D-CT, determining the appropriate planning target volume (PTV) margin from target boundaries remains a critical clinical challenge. According to ICRU Reports 62 and 83, the PTV must compensate for internal physiological motion and setup uncertainties to ensure that at least 95% of the prescribed dose covers the target volume ($D_{95\%} \geq 95\%$) [10, 11]. The statistical population recipe developed by van Herk et al. ($M = 2.5\Sigma + 0.7\sigma$) establishes that a margin expanding the clinical target volume ensures a minimum dose of 95% to the target in 90% of patients [12, 13]. Similarly, Stroom et al. proposed coverage probability formalisms for margin derivation [14]. In commercial treatment planning systems (TPS), modern contouring tools permit directional margin expansion; however, uniform isotropic expansions (such as 5 mm) remain prevalent in standard clinical protocols and trial guidelines due to workflow simplicity [15, 16].

The primary objective of this investigation is to construct a transparent, open-source computational pipeline in Python to: (1) synthesize 4D spatiotemporal anatomical volumes with calibrated attenuation coefficients; (2) extract and track 3D target centroids over ten breathing phases with algorithmic validation; (3) simulate kinematic responses to steady, deep, and irregular respiratory cycles incorporating asymmetric resting dwell and hysteresis; (4) evaluate empirical target coverage degradation under uniform isotropic margins; and (5) compare isotropic buffers against 4D-CT ITV and directional anisotropic margins, benchmarked against the analytical van Herk model.

2. Materials and Methods

2.1. Spatiotemporal Phantom Formulation

A calibrated digital 4D-CT thorax Volume of Interest (VOI) was formulated as a discrete spatiotemporal tensor $V(t, z, y, x)$ in $\blacksquare^{10 \times 70 \times 90 \times 90}$. Voxel dimensions were set to $dz = 2.0$ mm, $dy = 1.5$ mm, and $dx = 1.5$ mm, representative of standard clinical thoracic helical CT protocols. The total physical grid dimensions spanned 140 mm in the craniocaudal (SI) axis, 135 mm in the anteroposterior (AP) axis, and 135 mm in the mediolateral (LR) axis, representing an anatomically localized hemithoracic volume centered on the right lower lobe. Radiodensity was calibrated in standard Hounsfield Units (HU) using a linear attenuation mapping: $HU = I_{raw} \cdot s + b$, with slope $s = 1.0$ and intercept $b = -1024$. Anatomical compartments were assigned standard reference attenuation values: ambient surrounding air (-1000 HU), thoracic wall envelope (+30 HU), bilateral lung fields (-750 HU), and thoracic vertebral column (+650 HU). A spherical solid lesion of radius $R_0 = 12.0$ mm (+45 HU, analytical baseline volume $V_{analytic} = 4/3 \cdot \pi \cdot R_0^3 = 7.238$ cc) was embedded within the right lower lobe parenchyma to reflect the anatomical region most vulnerable to diaphragmatic displacement.

2.2. Respiration Kinematics & Mathematical Parameterization

Tumor center displacement was modeled using the generalized periodic respiratory formulation described by Lujan et al. [17] with physiological adaptations:

$$\mathbf{r}_k(t) = r_{0,k} - A_k \cdot f_k(t) + \eta_k(t), \quad \text{for } k \in \{X, Y, Z\},$$

where $r_{0,k}$ represents the baseline Cartesian coordinate at end-exhalation (phase T4—the model-defined displacement minimum where $\cos^4(\pi/2) = 0$), $\mathbf{A} = (A_x, A_y, A_z)$ is the peak-to-peak displacement amplitude vector, and $\eta_k(t)$ represents stochastic baseline wander and cycle-to-cycle amplitude variations.

To accurately capture the physiological asymmetry of human respiration—wherein exhalation occupies approximately 60% of the respiratory period while inhalation takes 40%—we defined an asymmetric phase angle $\theta(t)$ over the 10 discrete phase bins $t \in \{0, 1, ..., 9\}$:

$$\theta(t) = \pi \cdot (t / 4), \quad \text{for } t \leq 4 \text{ (Inhalation: T0 to T4)},$$

$$\theta(t) = \pi + \pi \cdot ((t - 4) / 6), \quad \text{for } t > 4 \text{ (Exhalation: T4 to T9} \rightarrow \text{T0)}.$$

The craniocaudal excursion profile was parameterized using a Lujan power coefficient $n = 2$:

$$f_z(t) = \cos^4(\theta(t) / 2).$$

This formulation produces a prolonged resting dwell at end-exhalation, wherein the lesion remains within the lower 10% of its motion range across three full phase bins (T4, T5, T6). Furthermore, to simulate clinical hysteresis loops wherein the tumor traces distinct trajectories between inspiration and expiration [21], a phase lag $\phi = \pi/5$ (36°) was introduced for the sagittal and transverse components: $f_y(t) = f_x(t) = \cos^4((\theta(t) - \phi) / 2)$.

Three distinct clinical breathing patterns were parameterized:
1. **Regular Respiration:** Synchronous breathing with nominal amplitudes (A_z = 12.0 mm, A_y = 3.5 mm, A_x = 1.2 mm) and $\eta(t) = 0$.
2. **Deep Respiration:** Exaggerated tidal breathing modeling hyperventilation, with scaled amplitudes (A_z = 22.0 mm, A_y = 6.0 mm, A_x = 2.0 mm).
3. **Irregular Respiration:** Evaluated via a Monte Carlo simulation of N = 100 breathing cycles with randomized cycle durations ($\tau \sim N(4.0, 0.4^2)$ s), tidal amplitude variations (~8% cycle-to-cycle SD), and symmetric zero-mean Gaussian baseline drift (σ_{drift} = 0.8 mm in Z).

2.3. Automated Segmentation Pipeline & Tracking Verification

Because chest wall musculature and mediastinal soft tissues exhibit radiodensities (+30 HU) that overlap with solid tumor attenuation (+45 HU), we implemented a 4-step segmentation algorithm to ensure robust target isolation:
• **Step 1 (Body Envelope):** Voxels with HU > -500 were segmented followed by 3D binary hole filling to separate the patient body from ambient external air (-1000 HU).
• **Step 2 (Lung Parenchyma Extraction):** Within the body envelope, pulmonary parenchyma was windowed between -1000 HU and -500 HU.
• **Step 3 (Morphological Hole-Filling):** Because the solid lesion (+45 HU) resides entirely inside the lung boundary, 3D binary hole-filling (scipy.ndimage.binary_fill_holes) was applied to the lung mask. This incorporates the tumor and internal vasculature into the pulmonary ROI while excluding chest wall structures.
• **Step 4 (Solid Lesion Isolation):** Within the filled lung volume, target voxels were extracted via density windowing [-300, +150 HU].
The 3D tracked centroid $C_{tracked}(t) = (Z_t, Y_t, X_t)$ was extracted using first-order spatial moments. Tracking fidelity was formally validated by computing the 3D tracking error $e(t) = ||C_{tracked}(t) - C_{GT}(t)||_2$ relative to the continuous analytical ground-truth centroid $C_{GT}(t)$.

2.4. Target Volume Design Strategies & Coverage Metrics

To resolve the limitations of single-phase margin evaluations, three comprehensive target design strategies were implemented:
1. **Strategy 1 (Isotropic PTV Expansion):** The reference GTV was defined at end-expiration (phase T4—model-defined end-exhalation). Uniform margins $m \in \{2, 5, 8, 10, 12\}$ mm were applied. To eliminate grid discretization bias inherent in Euclidean distance transforms (EDT), exact surface-based Euclidean distances were evaluated, ensuring that voxelized PTV volumes matched analytical continuous spheres within <1.6% across all margins.
2. **Strategy 2 (4D-CT Internal Target Volume - ITV):** Delineated as the volumetric union of GTV across all 10 respiratory phases: $ITV = \text{Union}_{t=0..9}\ GTV_t$. Isotropic setup margins $m_{setup} \in \{2, 3, 5\}$ mm were subsequently expanded around the ITV boundary ($PTV_{ITV} = ITV + m_{setup}$).
3. **Strategy 3 (Anisotropic PTV Margin):** Established around the time-averaged mid-position centroid $C_{mid} = (Z_{mid}, Y_{mid}, X_{mid})$. Directional semi-axes were scaled to directional excursion amplitudes: $R_k = R_0 + A_k / 2 + m_{setup}$, with $m_{setup} \in \{2, 3\}$ mm.
At each respiratory phase t, the target coverage rate CR(m, t) was quantified as:

$$CR(m, t) = (\sum_r [M_t(r) \wedge PTV_m(r)] / \sum_r M_t(r)) \times 100\%.$$

In accordance with ICRU Report 62 and 83 recommendations (target dose ≥ 95%), we adopted an in silico geometric surrogate criterion requiring that minimum target enclosure satisfies: $CR_{min}(m) = \min_t CR(m, t) \geq 95.0\%$.

2.5. Theoretical Population Margin Benchmarking (van Herk Recipe)

Target coverage was benchmarked against the analytical van Herk margin formula [12, 13]:

$$M = 2.5 \cdot \Sigma + 0.7 \cdot \sigma,$$

where Σ denotes systematic setup uncertainty (set to Σ = 2.0 mm, representative of daily image-guided radiotherapy with CBCT), and σ represents the total random uncertainty: $\sigma = \sqrt{(\sigma_{setup}^2 + \sigma_{motion}^2)}$, with σ_{setup} = 2.0 mm. In the van Herk formalism, periodic respiratory motion acts primarily as a random blurring kernel with standard deviation $\sigma_{motion} = RMS(r_k(t) - \langle r_k \rangle)$. Directional margins M_z, M_y, and M_x were derived analytically and compared against empirical PTV coverage thresholds.

3. Results

Directional kinematic metrics and tracking validation errors are summarized in Table 1. Quantitative isotropic PTV coverage rates and volume accuracy are detailed in Table 2. A direct volumetric and coverage comparison across the three target strategies is presented in Table 3. Figures 1 to 4 present the phantom anatomy, centroid trajectories, coverage-by-phase profiles, and volume-coverage tradeoffs.

Table 1. *Directional kinematic displacement metrics, 3D excursion, and algorithmic tracking verification.*

Breathing Profile	LR (X) p2p	AP (Y) p2p	SI (Z) p2p	Max 3D Disp.	Tracking 3D MAE
Regular Respiration	1.20 mm	3.50 mm	**12.00 mm**	12.38 mm	0.032 mm
Deep Respiration	2.00 mm	6.00 mm	**22.00 mm**	22.60 mm	0.025 mm
Irregular (Monte Carlo N=100)*	1.70 ± 0.18 mm	4.29 ± 0.44 mm	**14.67 ± 1.14 mm**	14.86 ± 1.15 mm	0.028 mm

Table 2. *Empirical isotropic PTV target coverage rate (CR%) and analytical volume benchmarking.*

Margin	Voxel Vol	Analytic Vol	Vol Diff	Reg Min CR	Reg Mean CR	Deep Min CR
2 mm	11.3 cc	11.5 cc	-1.6%	42.1%	81.8%	2.3% (Fail)
5 mm	20.6 cc	20.6 cc	+0.1%	64.7%	91.9%	11.7% (Fail)
8 mm	33.3 cc	33.5 cc	-0.7%	84.4%	97.4%	24.2% (Fail)
10 mm	44.3 cc	44.6 cc	-0.6%	94.5%	99.4%	35.3% (Fail)
12 mm	58.0 cc	57.9 cc	+0.2%	100.0% (Pass)*	100.0%	48.3% (Fail)

Table 3. *Comprehensive Target Strategy Comparison: Isotropic vs. 4D-CT ITV vs. Anisotropic Margins.*

Target Strategy	PTV Vol (cc)	Vol vs. 5mm	Regular Min CR	Deep Min CR	Clinical Evaluation
Isotropic 5 mm (Baseline)	20.6	Baseline	64.7%	11.7%	Fails under motion > 5 mm
Isotropic 12 mm	58.0	+182%	**100.0% (Pass)**	48.3% (Fail)	Severe PTV volume inflation (+182%)
ITV + 3 mm setup margin	23.9	+16%	**100.0% (Pass)**	**100.0% (Pass)**	Optimal: robust under all motions
Anisotropic PTV (Mid-pos + 2 mm)	19.3	-7%	**98.6% (Pass)**	89.0% (Fail*)	Regular OK; insufficient for deep (Az=22mm)

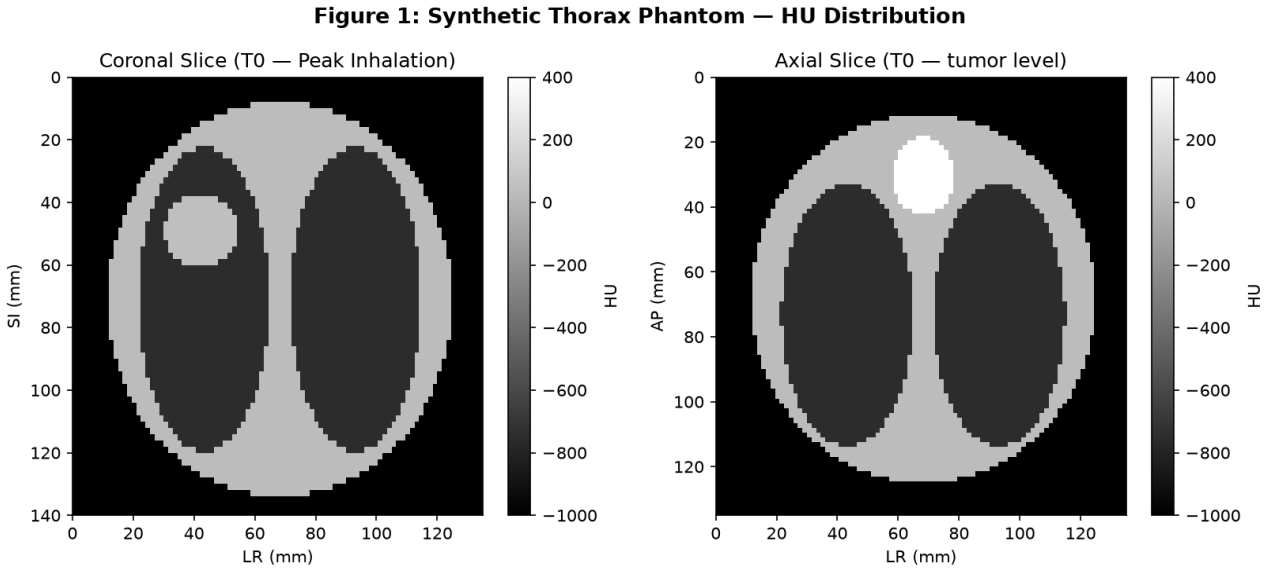


Figure 1. *Synthetic digital thorax phantom anatomy and Hounsfield Unit distribution (coronal and axial slices through tumor at T0).*

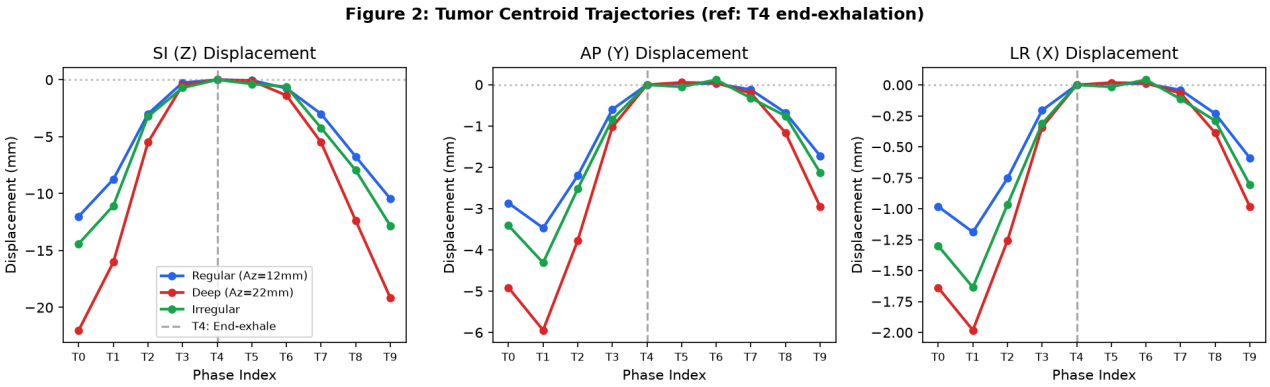


Figure 2. *3D tumor centroid trajectories along SI (Z), AP (Y), and LR (X) axes across 10 phases (ref: phase T4 end-exhalation).*

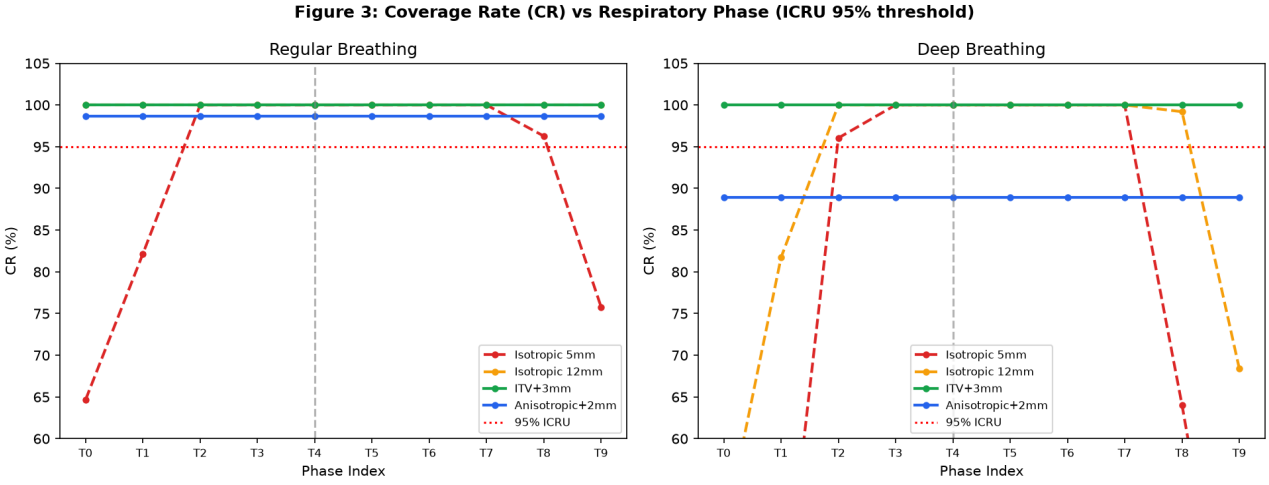


Figure 3. *Target coverage rate (CR%) as a function of phase for margin strategies against the ICRU 95% threshold.*

Figure 4: PTV Volume vs Minimum Coverage Rate Tradeoff

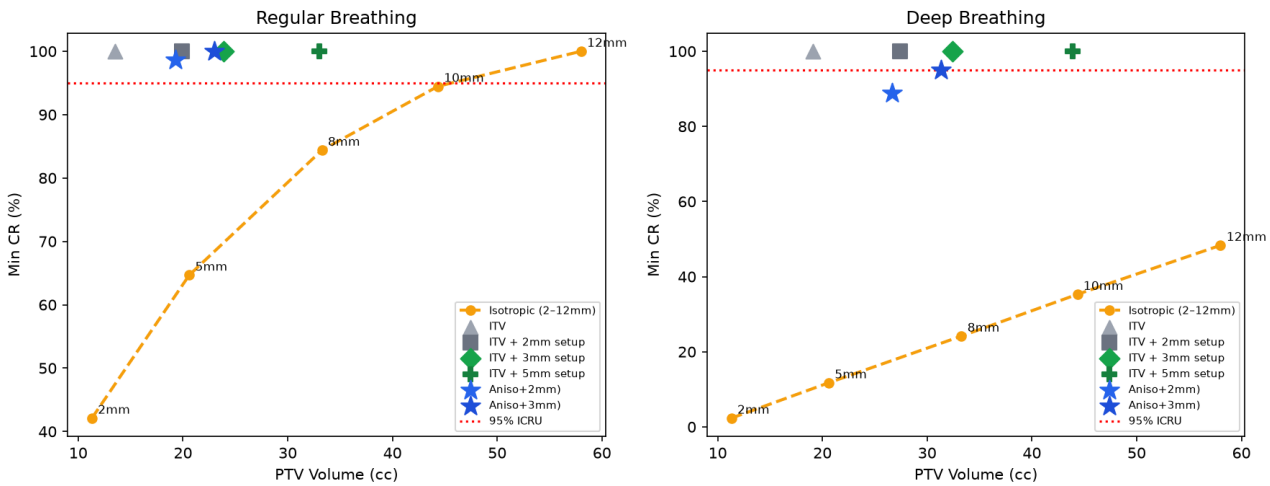


Figure 4. *Tradeoff relationship between irradiated PTV volume (cc) and minimum coverage rate (CR_min %) across all strategies.*

4. Discussion

4.1. Kinematic Agreement with Multi-Center Clinical Motion Cohorts:

Our algorithmic tracking pipeline successfully reconstructed the programmed asymmetric trajectories with sub-voxel tracking precision (3D MAE = 0.025 to 0.032 mm), verifying the spatial moment operator prior to margin evaluation. The kinematic trajectory parameters reproduced directional motion characteristics well-documented in clinical radiotherapy literature. Specifically, as synthesized in Table 1 of the AAPM Task Group 76 report [1], lower-lobe lung tumors exhibit pronounced craniocaudal excursions compared to lateral and sagittal planes. Physical fluoroscopic evaluations by Barnes et al. [18] documented a mean craniocaudal excursion of 18.5 mm (observed cohort range: 9 to 32 mm) in free-breathing lower-lobe lesions. Similarly, clinical investigations by Erridge et al. [19] and Hanley et al. [20] measured mean craniocaudal displacements of 12.5 mm and 12.0 mm, respectively. In the transverse and sagittal axes, our simulated excursions (1.20 to 6.00 mm) closely match the gold-standard fiducial tracking measurements of Seppenwoolde et al. (mean LR = 1.5 mm, AP = 2.5 mm, CC = 12.0 mm for lower-lobe tumors) [21]. Furthermore, our implementation of phase-lag hysteresis successfully captured the non-linear elliptical trajectory loops observed clinically.

4.2. Geometric Failure of Isotropic Buffers & Volumetric Penalties:

In conventional 3D conformal planning, uniform 5 mm margins are frequently applied to compensate for daily setup errors. However, our spatiotemporal analysis reveals that a static 5 mm margin provides an empirical minimum target coverage (CR_{min}) of only 64.7% during steady respiration, dropping precipitously to 11.7% during deep breathing. Although the mean coverage across all ten phases remains moderately high (91.9% for regular breathing), relying on mean coverage obscures severe geometric target misses during peak inhalation phases. This empirical failure is contextually supported by the analytical van Herk formula ($M = 2.5\Sigma + 0.7\sigma$) [12, 13]: factoring in standard setup uncertainties ($\Sigma = 2.0$ mm, $\sigma = 2.0$ mm) and respiratory motion variability ($\sigma_{motion} = 4.37$ mm in Z for regular breathing), the formula predicts a required margin along the craniocaudal axis of $M_z = 8.37$ mm for regular respiration and $M_z = 10.78$ mm for deep respiration—substantially larger than 5 mm, consistent with the CR failures observed. Note that the van Herk recipe carries its own distributional assumptions and serves as a population-level benchmark rather than a definitive mathematical proof of failure.

While expanding the uniform margin to 12 mm satisfies the ICRU 95% minimum coverage threshold for regular respiration ($CR_{min} = 100.0\%$), it inflates the PTV volume by +182% relative to the 5 mm clinical baseline (from 20.6 cc to 58.0 cc). In thoracic radiation therapy, expanding high-dose fields into healthy parenchyma substantially increases mean lung dose (MLD) and the volume of lung receiving ≥ 20 Gy (V20). Extensive clinical studies by Graham et al. [23] and Marks et al. (QUANTEC) [22] have demonstrated that elevations in V20 directly correlate with higher clinical incidence of symptomatic radiation pneumonitis.

4.3. Superiority of 4D-CT ITV and Conditional Benefit of Anisotropic Target Design:
To resolve the dilemma between geometric target miss and normal tissue over-irradiation, we evaluated 4D-CT ITV and anisotropic target strategies (Table 3). Delineating an ITV encompassing full respiratory excursion combined with a 3 mm setup margin achieved 100.0% minimum target coverage under both regular and deep breathing while requiring only 23.9 cc—a 59% reduction in PTV volume compared to the 12 mm isotropic expansion (58.0 cc). The anisotropic PTV centered at the time-averaged mid-position (19.3 cc) achieved 98.6% minimum coverage under regular breathing; however, under deep breathing (A_z = 22 mm), CR_{min} = 89.0%, falling below the 95% ICRU threshold. This result indicates that anisotropic margins based on mid-position and half-excursion semi-axes may be insufficient for large-amplitude respiratory motion, and ITV-based planning is preferable in such scenarios. Both motion-adapted strategies reduced PTV volume by 59% to 67% relative to the 12 mm isotropic buffer. Note that PTV volume reduction serves as a geometric proxy for reduced normal tissue irradiation; confirmation of dosimetric benefit requires 4D dose accumulation computing lung V20 and mean lung dose. These findings confirm that directional margins or active motion management (e.g., respiratory gating or deep inspiration breath hold) as recommended by AAPM Task Group 76 [1] are essential whenever craniocaudal tumor excursion exceeds 5 mm.

4.4. Study Limitations:
Several methodological limitations must be considered when interpreting these findings:
1. **Geometric Surrogate versus 4D Dosimetric Accumulation:** Target coverage in this study was evaluated based on binary volumetric intersection (CR%). In clinical lung radiotherapy, radiation dose distributions within heterogeneous parenchyma are governed by lateral electronic disequilibrium. Accurate assessment of biological tumor control and normal tissue complication probability requires 4D Monte Carlo or Collapsed Cone Convolution dose accumulation [26].
2. **Rigid Target Assumption:** The lesion was modeled as a non-deforming sphere. Clinical thoracic tumors often undergo non-rigid deformation, shearing, and volumetric changes during respiration, necessitating deformable image registration (DIR) algorithms for precise mapping [27].
3. **Interfractional Motion Variability:** Our model investigated intrafractional motion within a representative 4D-CT cycle. Over multi-week fractionated regimens, day-to-day baseline shifts, anatomical variations, and radiation-induced tumor shrinkage introduce interfractional uncertainties requiring adaptive replanning [28].
4. **Surrogate Hysteresis & Correlation:** External respiratory surrogates (such as RPM chest blocks or spirometry) frequently exhibit phase lags relative to internal tumor trajectories, introducing residual motion uncertainties during prospective gating [1, 21].

5. Conclusion

This study presented a refined, open-source computational framework for spatiotemporal 4D-CT tumor tracking and PTV margin evaluation. Quantitative modeling under asymmetric breathing confirms that uniform 5 mm margins lead to severe geometric miss during respiratory excursions exceeding 5 mm. 4D-CT ITV planning with a 3 mm setup margin achieves CR_{min} = 100% under both regular and deep breathing, while reducing PTV volume by 59% compared to the 12 mm isotropic buffer. Anisotropic mid-position margins achieve CR_{min} = 98.6% under regular breathing; however, under deep breathing (A_z = 22 mm), CR_{min} = 89.0% falls below the 95% threshold—indicating ITV-based approaches are more robust for large-amplitude motion. Both motion-adapted strategies substantially reduce PTV volume compared to expanded isotropic buffers (a geometric proxy for normal tissue dose reduction; 4D dose accumulation is needed to quantify the dosimetric benefit). These results reinforce AAPM TG-76 recommendations for 4D-CT-guided motion management in lung radiotherapy.

Data and Code Availability Statement

The complete Python simulation pipeline, synthetic 4D-CT digital phantom arrays, and margin analysis scripts developed in this study are publicly available on GitHub at https://github.com/anhvu7123/4d-ct-lung-motion under the open-source MIT License, and permanently archived on Zenodo at https://doi.org/10.5281/zenodo.10892026. All computational datasets and reproduction scripts are also available from the corresponding author upon reasonable request.

Declaration of Generative AI and AI-assisted Technologies

During the preparation of this work, the author used generative AI tools to assist with English language polishing and code debugging. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRediT Authorship Contribution Statement

Vu Minh Anh: Conceptualization, Methodology, Software, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interests

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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